

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Error Analysis Task

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1504

COURSE OUTCOME COVERED

CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN. (BL5)*

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 25

Code of Conduct

I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks Error Analysis Task

1. A Hadoop job repeatedly fails because input splits are uneven and reducers remain idle. Identify the likely design error and propose corrective actions.

Error Analysis Problem

The Hadoop job repeatedly fails because **input splits are uneven**, causing some mapper tasks to receive much more data than others. As a result, some reducers remain idle while waiting for the slow tasks to finish.

Likely Design Error

The main design error is **poor input-split and partition design**. Unequal data distribution creates a **data-skew problem**, resulting in unbalanced mapper workloads and inefficient reducer utilization.

Corrective Actions

- Create **more evenly sized input splits**.
- Use an appropriate HDFS block size for the workload.
- Balance data across DataNodes.
- Use a suitable **partitioner** to distribute intermediate data evenly among reducers.
- Increase or adjust the number of reducers according to the workload.
- Monitor mapper and reducer execution times to identify remaining data skew.

Conclusion

The problem can be resolved by **balancing input splits and intermediate data distribution**, ensuring that mapper and reducer tasks receive workloads more evenly and improving overall Hadoop job performance.

2. An HDFS deployment shows frequent data unavailability after a single node failure. Analyze the probable replication or NameNode design error.

Error Analysis

Problem

The HDFS deployment experiences **frequent data unavailability after a single node failure**.

Probable Design Error

The likely error is an **insufficient replication factor** or an incorrect **NameNode design**. If data blocks have too few replicas, failure of one DataNode can make the required data unavailable. A single, non-redundant NameNode can also create a single point of failure.

Corrective Actions

- Set an appropriate HDFS **replication factor**, such as 3.
- Distribute replicas across different DataNodes and racks.
- Use **NameNode High Availability** with active and standby NameNodes.
- Monitor under-replicated blocks and automatically restore missing replicas.
- Regularly check DataNode and NameNode health.

Conclusion

The problem can be corrected by ensuring **adequate block replication and NameNode high availability**. This provides continued access to data even when a DataNode or NameNode component fails.

3. A data ingestion workflow using ETL and Apache NiFi creates duplicate records in the analytics store. Identify the likely causes and recommend fixes.

Error Analysis

Problem

The data ingestion workflow using **ETL and Apache NiFi** creates duplicate records in the analytics store. **Likely Causes**

The main causes may include:

- The same source data is processed more than once.
- NiFi processors may retry a flow without proper duplicate checking.
- ETL jobs may lack **unique record identification**.
- Incorrect flow design may route the same data to the analytics store multiple times.

Corrective Actions

- Assign a **unique ID** to each incoming record.
- Use NiFi flow control to prevent duplicate processing.
- Configure proper success and failure relationships between processors.
- Apply **deduplication checks** before inserting records.
- Use database constraints such as unique keys where applicable.
- Monitor and log processed records to identify repeated ingestion.

Conclusion

The duplication problem can be resolved by implementing **unique record identification, proper NiFi flow control, and deduplication checks** before data reaches the analytics store. This ensures consistent and reliable data ingestion.

4. A YARN cluster suffers from resource starvation when multiple users submit jobs together. Analyze the scheduling error and suggest a better resource management approach.

Error Analysis

Problem

The YARN cluster suffers from **resource starvation when multiple users submit jobs simultaneously**.

Scheduling Error

The likely error is **poor resource allocation or an unsuitable scheduling policy**. One or more jobs may consume excessive CPU or memory, preventing other users' applications from receiving sufficient resources.

Corrective Actions

- Configure a suitable **YARN scheduler** such as Capacity Scheduler or Fair Scheduler.
- Create separate queues for different users or workloads.
- Set **resource limits and minimum allocations** for each queue.
- Give important jobs appropriate scheduling priority.
- Prevent a single user or application from consuming all cluster resources.
- Monitor CPU and memory utilization regularly.

Conclusion

The problem can be solved through **fair resource allocation, queue-based scheduling, and resource limits**. This prevents resource starvation and ensures that multiple users can execute their jobs efficiently.

5. A backup strategy for distributed storage restores outdated data after recovery. Identify the probable design flaw in the replication or backup approach.

Error Analysis

Problem

The backup strategy for distributed storage restores **outdated data after recovery**.

Probable Design Flaw

The likely flaw is an **incorrect replication or backup strategy**. The backup may not be updated regularly, or recovery may be using an older backup copy instead of the latest valid version.

Corrective Actions

- Schedule **regular and incremental backups**.
- Maintain timestamps or version information for backup copies.

- Verify backup completion and data consistency.
- Use the latest valid backup during recovery.
- Monitor replication status and ensure replicas are synchronized.
- Periodically perform recovery tests to verify backup reliability.

Conclusion

The problem can be corrected by maintaining **up-to-date backups, synchronized replicas, and proper version tracking**. Regular recovery testing ensures that the latest valid data can be restored when required.

Error Analysis Task Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Identification of Error	30%	Accurately identifies the root technical issue	Identifies most of the issue	Partially identifies issue	Fails to identify issue
Cause Analysis	20%	Explains root cause with strong technical reasoning	Reasonable cause analysis	Basic cause analysis	Weak analysis
Corrective Action	25%	Proposes effective and feasible corrections	Reasonable corrections	Basic corrections	Ineffective corrections
Use of Hadoop / Integration Concepts	15%	Applies relevant concepts appropriately	Mostly appropriate application	Partial application	Incorrect application
Clarity	10%	Clear and direct explanation	Generally clear	Somewhat unclear	Poorly presented